

Campus Notification Engine

Reach 30,000 students fast — without turning every campus event into notification spam.

THE CARD

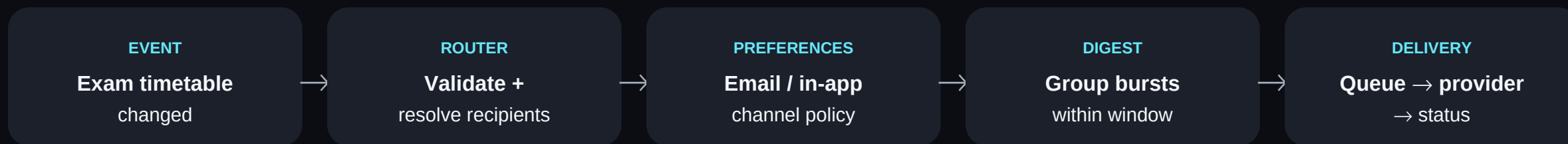
Exam timetable changes must reach students quickly, respect channel preferences, group bursts into digests, and retry failures without duplicates.

OUR THESIS

A notification engine should optimize attention and trust — not just delivery speed.

From campus event to reliable delivery

One event becomes preference-aware, deduplicated delivery work.



3 KILLER TESTS

- 10 events within five minutes → one digest.
- Muted email → in-app only; no email send.
- Failed send → retry without a duplicate logical notification.

RELIABILITY PRINCIPLES

- Notification identity is separate from delivery attempts.
- Database constraints + idempotency prevent duplicate logical sends.
- Failures become explicit states rather than silent drops.

Fail-closed webhook verification

Trust an inbound delivery callback only after proving its source.

INBOUND PROVIDER WEBHOOK

Webhook



VERIFY SIGNATURE



VALID
PROCESS



INVALID / NONE
REJECT

Unsupported verification must never silently become “verified”.

WHY IT MATTERS

The reverse-engineering analysis identified a fail-open trust boundary in inbound provider callbacks.

- Invalid signatures are rejected instead of updating delivery state.
- Unsupported verification is rejected or explicitly treated as unverified.
- The fix protects delivery records from forged “delivered” / “failed” callbacks.
- This is a true Fix: it corrects existing behavior rather than adding a new capability.

FIX

Trust the event only after proving its source.

Intelligent Focus Mode

Don't just delay notifications. Understand what changed while the student was away.

1 FOCUS

Student starts a
2-hour focus window.

2 FILTER

Normal events wait;
critical events bypass.

3 CATCH UP

Related events become
one actionable summary.

WHY IT'S NEW

- Time-bounded focus
- Critical bypass
- Event correlation
- Actionable catch-up
- Redundant follow-up suppression

NOT A DIGEST

Digest = messages grouped.
Focus Mode = meaning grouped.

EXAMPLE • EXAM CHANGE

5 related notifications → **1 actionable catch-up**

“DBMS exam changed — 10:00 AM → 2:00 PM. Room AB2-301 → AB3-204. Check the new room.”

Critical event? Bypass Focus Mode and send immediately.

Clean-room implementation & demo story

Built from the frozen seven-document specification, then implemented as a fresh system.

F

Web UI

inbox • preferences • focus mode

HTTP API

events • notifications • preferences

Notification Engine

routing • digest • focus • idempotency

Persistence

events • messages • delivery state

Providers

email + in-app delivery

JUDGE-READY CHECKLIST

- 3 Killer Tests are the first acceptance gate.
- Fix: fail-closed provider webhook verification.
- Differentiator: Intelligent Focus Mode.
- README.md, SUBMISSION.md, deck.pdf and .env.example at repo root.
- Seven reverse-engineering docs remain under docs/.
- No .env or node_modules committed; the rebuild uses the docs as its implementation contract.